

# UNIT III – GRAMMARS AND APPLICATIONS – 30 MCQs

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\* Indicates required question

## MCQ

15. Which of the following is NOT normally a step in simplifying a CFG? \*

- ☐ Removing useless symbols
- ☐ Removing  $\epsilon$ -productions
- ☐ Removing unit productions
- ☒ Adding new DFA states

6. Which machine recognizes Context-Free Languages? \*

- ☐ Finite Automaton
- ☒ Pushdown Automaton
- ☐ Turing Machine only
- ☐ Linear Bounded Automaton



8. Which of the following is a valid CFG production? \*

- ☐  $AB \rightarrow a$
- ☒  $A \rightarrow aB$
- ☐  $ab \rightarrow AB$
- ☐  $aB \rightarrow bA$

3. According to the Chomsky hierarchy, Type-0 grammars are called: \*

- ☐ Context-Free Grammars
- ☐ Regular Grammars
- ☒ Unrestricted Grammars
- ☐ Context-Sensitive Grammars

4. Type-2 grammars in the Chomsky hierarchy are: \*

- ☒ Context-Free Grammars
- ☐ Regular Grammars
- ☐ Context-Sensitive Grammars
- ☐ Unrestricted Grammars

14. The leaf nodes of a parse tree, when read from left to right, give: \*

- ☐ The grammar
- ☐ The derivation sequence
- ☒ The generated string
- ☐ The set of non-terminals



11. In a rightmost derivation, which symbol is replaced first? \*

- ☐ Leftmost non-terminal
- ☐ Leftmost terminal
- ☒ Rightmost non-terminal
- ☐ Start symbol only

25. Context-Free Languages are closed under: \*

- ☒ Union
- ☐ Intersection
- ☐ Complement
- ☐ Difference

30. In the Pumping Lemma for CFLs, a string  $z$  is typically represented as: \*

- ☒  $z=uvwxy$
- ☐  $z=uvwx$
- ☐  $z=abcde$  only
- ☐  $z=xyz$

20. In Chomsky Normal Form, a production of the form  $A \rightarrow a$  contains: \*

- ☒ One terminal
- ☐ Two terminals
- ☐ One non-terminal
- ☐ Two non-terminals



17. A production of the form  $A \rightarrow B$ , where both A and B are non-terminals, is called: \*

- ☐ Null production
- ☒ Unit production
- ☐ Terminal production
- ☐ Binary production

13. In a parse tree, the root node represents: \*

- ☐ A terminal symbol
- ☐ The first production
- ☒ The start symbol
- ☐ The last terminal

22. In Greibach Normal Form, productions generally have the form: \*

- ☐  $A \rightarrow BC$
- ☒  $A \rightarrow a\alpha$
- ☐  $A \rightarrow \epsilon$  only
- ☐  $A \rightarrow AB$



23. In Greibach Normal Form, the first symbol on the right-hand side must be: \*

- ☐ A non-terminal
- ☒ A terminal
- ☐ The start symbol
- ☐ An epsilon symbol

12. A parse tree is also called a: \*

- ☐ Transition tree
- ☒ Derivation tree
- ☐ State tree
- ☐ Finite tree

5. A production of the form  $A \rightarrow \alpha$ , where A is a single non-terminal, is characteristic of: \*

- ☐ Type-0 grammar
- ☐ Type-1 grammar
- ☒ Type-2 grammar
- ☐ Type-3 grammar



28. The Pumping Lemma for CFLs is primarily used to: \*

- ☐ Construct a DFA
- ☒ Prove that certain languages are not context-free
- ☐ Convert CFG to CNF
- ☐ Minimize a PDA

2. Which of the following is a component of a grammar? \*

- ☒ Production rules
- ☐ Transition function only
- ☐ Stack alphabet only
- ☐ Final states

7. In a CFG, the symbol S generally represents: \*

- ☒ Start symbol
- ☐ State symbol
- ☐ Stack symbol
- ☐ Separator

18. Which normal form requires productions mainly of the form  $A \rightarrow BC$  or  $A \rightarrow a$ ? \*

- ☐ Greibach Normal Form
- ☒ Chomsky Normal Form
- ☐ Regular Normal Form
- ☐ Reduced Normal Form



26. Context-Free Languages are NOT generally closed under: \*

- ☒ Union
- ☐ Concatenation
- ☐ Kleene star
- ☐ Intersection

27. Which operation preserves Context-Free Languages? \*

- ☐ Complement
- ☐ Intersection of two CFLs
- ☒ Concatenation
- ☐ Difference

29. If L is a context-free language, the Pumping Lemma states that sufficiently long strings can be divided into: \*

- ☐ Two parts
- ☐ Three parts
- ☒ Four parts
- ☐ Five parts



10. In a leftmost derivation, which symbol is replaced first? \*

- ☒ Rightmost terminal
- ☐ Leftmost non-terminal
- ☐ Rightmost non-terminal
- ☐ Leftmost terminal

19. In Chomsky Normal Form, the production  $A \rightarrow BC$  contains: \*

- ☐ Two terminals
- ☒ Two non-terminals
- ☐ One terminal and one non-terminal
- ☐ Three non-terminals

9. A derivation in a grammar is a sequence of: \*

- ☐ State transitions
- ☐ Rewriting steps using production rules
- ☐ Stack operations
- ☒ DFA transitions

24. Which of the following is an application of Context-Free Grammars? \*

- ☒ Syntax specification of programming languages
- ☐ IP address assignment only
- ☐ Hardware manufacturing
- ☐ Image compression only





16. A production of the form  $A \rightarrow \epsilon$  is called: \*

- ☐ Unit production
- ☒ Null or epsilon production
- ☐ Terminal production
- ☐ Binary production

21. In Chomsky Normal Form, the start symbol may produce: \*

- ☐ Only two terminals
- ☒  $\epsilon$ , under the usual exception when  $\epsilon$  belongs to the language
- ☐ Any arbitrary string
- ☐ Only another start symbol

1. A grammar is formally represented as: \*

- ☒  $G=(Q,\Sigma,\delta,q_0,F)$
- ☐  $G=(V,T,P,S)$
- ☐  $G=(V,E,\delta,S)$
- ☐  $G=(S,V,T,Q)$

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